

Benjamin Choon Heng Lee

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Google Scholar

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RESEARCH INTERESTS

My research lies at the intersection of human-computer interaction, data visualization, and augmented and virtual reality. I design, implement, and evaluate interactive systems that leverage emerging technologies to support data understanding, engagement, and decision-making. I have a deep interest in investigating how established practices can be augmented through spatial computing techniques, across both computational settings (Immersive Analytics) and physical tasks (Situated Analytics). I also develop tools and grammars that enable to highly expressive and interactive 3D data visualizations, representations, and animations.

RESEARCH EXPERIENCE

JPMorgan Chase & Co.

New York, NY, USA

Senior Associate Research Scientist at *Global Technology Applied Research*

June 2024 - Present

- I conduct observational studies, interviews, and collaborate with internal stakeholders to understand existing workplace practices and identify opportunities for improvement through emerging technologies
- I lead the design and development of user-centered prototypes that integrate AI/ML techniques including computer vision, natural user interactions, and real-time data visualization to promote collaboration, engagement, and learning outcomes
- I design, develop, and maintain an open-source 3D data visualization toolkit used by top researchers and validated through expert evaluations and interviews
- I present user-centered research, design work, and fully functional prototypes at firmwide events with up to over 1000+ attendees, and deliver executive briefings to internal and external leadership including C-suite executives
- Manager: Blair MacIntyre

University of Stuttgart

Stuttgart, Baden-Württemberg, Germany

Postdoctoral Researcher at *Visualization Research Center (VISUS)*

February 2023 - April 2024

- I led the design and analysis of a mixed-methods usability study investigating interaction techniques to support computational chemists' scientific workflows using augmented reality
- I designed, developed, and evaluated visual highlighting techniques for in-situ decision-making with objects in simulated virtual reality environments
- I led the iterative ideation and creation of design spaces and patterns that establish standardized terminology and visual idioms in the fields of situated visualization and motion guidance
- I co-supervised and mentored doctoral students (ongoing as external)
- Manager: Prof. Dieter Schmalstieg

Microsoft Research

Redmond, WA, USA

Research Intern at *Visualization and Interactive Data Analysis Group*

June 2019 - September 2019

- I led the ideation, design, development, and qualitative usability evaluation of a novel approach for communicating data to general audiences by eliciting visceral emotional responses in virtual reality
- Mentors: Dr. Dave Brown, Dr. Steven Drucker, Prof. Bongshin Lee

Monash University

Melbourne, Victoria, Australia

Doctoral Researcher at *Data Visualization and Immersive Analytics Lab*

February 2019 - January 2023

- I led the design and development of novel systems in virtual and augmented reality for collaborative visual analytics, employing qualitative approaches to evaluate their usability with dyads and triads

- I led the conceptualization of a design space that defines animated interactive transitions between 2D and 3D data visualizations, and subsequently formalized this design space as a visualization grammar and functional prototype
- Advisors: Prof. Tim Dwyer, A/Prof. Bernhard Jenny, A/Prof. Maxime Cordeil, Dr. Arnaud Prouzeau

EDUCATION

Monash University

Ph.D. in Immersive Analytics

Melbourne, Victoria, Australia

February 2019 – January 2023

- Thesis: *Surfaces and Spaces in Immersive Analytics*
- Advisors: Prof. Tim Dwyer, A/Prof. Bernhard Jenny, A/Prof. Maxime Cordeil, Dr. Arnaud Prouzeau
- Examiners: Prof. Harald Reiterer, Prof. Luciana Nedel

Monash University

Bachelor of Informatics and Computation Advanced (First Class Honours)

Melbourne, Victoria, Australia

2015 – 2018

- Thesis: *Heterogeneous Mixed-Reality Display Environments for Immersive Visual Analytics*
- Advisors: Prof. Tim Dwyer, A/Prof. Bernhard Jenny, A/Prof. Maxime Cordeil

PUBLICATIONS

The following lists exclude short papers, posters, workshops, and extended abstracts. Please see my Google Scholar page for a complete list.

Journal Articles

* Denotes equal contribution

- 2026 [Hybrid User Interfaces: Past, Present, and Future of Complementary Cross-Device Interaction in Mixed Reality](#)
Sebastian Hubenschmid*, Marc Satkowski*, Johannes Zagermann*, Julián Méndez*, Niklas Elmqvist, Steven Feiner, Tiare Feuchtnr, Jens Emil Grønbaek, **Benjamin Lee**, Dieter Schmalstieg, Raimund Dachsel, and Harald Reiterer
Under review at IEEE Transactions on Visualization and Computer Graphics
DOI: 10.48550/ARXIV.2509.05491
- 2026 [Design Considerations for Visualization Transitions of 3D Spatial Data in Hybrid AR-Desktop Environments](#)
Yucheng Lu, Tobias Rau, **Benjamin Lee**, Andreas Köhn, Michael Sedlmair, Christian Sandor, and Tobias Isenberg
To appear in Computer Graphics Forum
DOI: 10.48550/ARXIV.2506.22250
- 2026 [Situating Brushing and Linking in Virtual and Augmented Reality](#)
Carlos Quijano-Chavez, **Benjamin Lee**, Nina Doerr, Wolfgang Büschel, Michael Sedlmair, and Dieter Schmalstieg
To appear in IEEE Transactions on Visualization and Computer Graphics (IEEE VR 2026)
- 2025 [Desktop versus VR for collaborative sensemaking](#)
Ying Yang, Tim Dwyer, Zachari Swiecki, **Benjamin Lee**, Michael Wybrow, Maxime Cordeil, Teresa Wulandari, Bruce H. Thomas, and Mark Billinghurst
Frontiers in Virtual Reality
DOI: 10.3389/frvir.2025.1570383
- 2024 [Visual Highlighting for Situated Brushing and Linking](#)
Nina Doerr, **Benjamin Lee**, Katarina Baricova, Dieter Schmalstieg, and Michael Sedlmair
Computer Graphics Forum
DOI: 10.1111/cgf.15105
- 2024 [Design Patterns for Situated Visualization in Augmented Reality](#)
Benjamin Lee, Michael Sedlmair, and Dieter Schmalstieg
IEEE Transactions on Visualization and Computer Graphics (IEEE VIS 2023)
DOI: 10.1109/TVCG.2023.3327398

- 2024 [PerSiVal: On-Body AR Visualization of Biomechanical Arm Simulations](#)
Xingyao Yu, David Rosin, Johannes Kässinger, **Benjamin Lee**, Frank Dürr, Christian Becker, Oliver Röhrle, and Michael Sedlmair
IEEE Computer Graphics and Applications
doi: 10.1109/MCG.2024.3494598
- 2024 [Lights, Headset, Tablet, Action: Exploring the Use of Hybrid User Interfaces for Immersive Situated Analytics](#)
Xiaoyan Zhou*, **Benjamin Lee***, Francisco R. Ortega, Anil Ufuk Batmaz, and Yalong Yang
Proceedings of the ACM on Human-Computer Interaction (ACM ISS 2024)
doi: 10.1145/3698147
- 2022 [Towards immersive collaborative sensemaking](#)
Ying Yang, Tim Dwyer, Michael Wybrow, **Benjamin Lee**, Maxime Cordeil, Mark Billingham, and Bruce H. Thomas
Proceedings of the ACM on Human-Computer Interaction (ACM ISS 2022)
doi: 10.1145/3567741
- 2021 [Data Visceralization: Enabling Deeper Understanding of Data Using Virtual Reality](#)
Benjamin Lee, Dave Brown, Bongshin Lee, Christophe Hurter, Steven Drucker, and Tim Dwyer
IEEE Transactions on Visualization and Computer Graphics (IEEE VIS 2020)
doi: 10.1109/TVCG.2020.3030435
 **Honorable Mention Award (Top 5% of Submissions)**
- 2021 [Shared Surfaces and Spaces: Collaborative Data Visualisation in a Co-located Immersive Environment](#)
Benjamin Lee, Xiaoyun Hu, Maxime Cordeil, Arnaud Prouzeau, Bernhard Jenny, and Tim Dwyer
IEEE Transactions on Visualization and Computer Graphics (IEEE VIS 2020)
doi: 10.1109/TVCG.2020.3030450
- 2021 [The MADE-Axis: A Modular Actuated Device to Embody the Axis of a Data Dimension](#)
Jim Smiley, **Benjamin Lee**, Siddhant Tandon, Maxime Cordeil, Lonni Besançon, Jarrod Knibbe, Bernhard Jenny, and Tim Dwyer
Proceedings of the ACM on Human-Computer Interaction (ACM ISS 2021)
doi: 10.1145/3488546
 **Honorable Mention Award (Top 5% of Submissions)**

Conference Proceedings

* Denotes equal contribution

- 2026 [S-TIER: Situated-Traceable Insights in Extended Reality for Hybrid Crisis Management](#)
Feiyu Lu*, Cheng Yao Wang*, Leonardo Pavanatto, Fannie Liu, David Saffo, **Benjamin Lee**, Mengyu Chen, and Blair MacIntyre
Under submission to IEEE International Symposium on Mixed and Augmented Reality
- 2026 [ARCurrencyCrittters: an Augmented Reality Game for Supporting Historically Marginalized Youth's Financial Literacy](#)
Erica Principe Cruz, David Saffo, Leonardo Pavanatto, **Benjamin Lee**, Feiyu Lu, Youngwook Do, Mengyu Chen, Cheng Yao Wang, Blair MacIntyre, and Fannie Liu
Under review at ACM Interaction Design and Children Conference
- 2026 [Anu.js: Accelerating Web-based Immersive Analytics](#)
David Saffo*, **Benjamin Lee***, Feiyu Lu, Cheng Yao Wang, and Blair MacIntyre
To appear in ACM CHI Conference on Human Factors in Computing Systems
- 2025 [Examining the Effects of Immersive and Non-Immersive Presenter Modalities on Engagement and Social Interaction in Co-located Augmented Presentations](#)
Matt Gottsacker, Mengyu Chen, David Saffo, Feiyu Lu, **Benjamin Lee**, and Blair MacIntyre
ACM CHI Conference on Human Factors in Computing Systems
doi: 10.1145/3706598.3713346
- 2025 [Traversing Dual Realities: Investigating Techniques for Transitioning 3D Objects between Desktop and Augmented Reality Environments](#)
Tobias Rau, Tobias Isenberg, Andreas Koehn, Michael Sedlmair, and **Benjamin Lee**
ACM CHI Conference on Human Factors in Computing Systems
doi: 10.1145/3706598.3713949
 **Best Paper Award (Top 1% of Submissions)**

- 2024 [Design Space of Visual Feedforward And Corrective Feedback in XR-Based Motion Guidance Systems](#)
Xingyao Yu, **Benjamin Lee**, and Michael Sedlmair
ACM CHI Conference on Human Factors in Computing Systems
DOI: 10.1145/3613904.3642143
- 2023 [Deimos: A Grammar of Dynamic Embodied Immersive Visualisation Morphs and Transitions](#)
Benjamin Lee, Arvind Satyanarayan, Maxime Cordeil, Arnaud Prouzeau, Bernhard Jenny, and Tim Dwyer
ACM CHI Conference on Human Factors in Computing Systems
DOI: 10.1145/3544548.3580754
- 2022 [A Design Space For Data Visualisation Transformations Between 2D And 3D In Mixed-Reality Environments](#)
Benjamin Lee, Maxime Cordeil, Arnaud Prouzeau, Bernhard Jenny, and Tim Dwyer
ACM CHI Conference on Human Factors in Computing Systems
DOI: 10.1145/3491102.3501859
🏆 Honorable Mention Award (Top 5% of Submissions)
- 2022 [Tangible Globes for Data Visualisation in Augmented Reality](#)
Kadek Ananta Satriadi, Jim Smiley, Barrett Ens, Maxime Cordeil, Tobias Czauderna, **Benjamin Lee**, Ying Yang, Tim Dwyer, and Bernhard Jenny
ACM CHI Conference on Human Factors in Computing Systems
DOI: 10.1145/3491102.3517715

AWARDS

- **Best Paper Award (Top 1% of Submissions)** 2025
ACM CHI Conference on Human Factors in Computing Systems
- **Vice-Chancellor's Commendation For Thesis Excellence Award** 2023
Monash University
- **Honorable Mention Award (Top 5% of Submissions)** 2022
ACM CHI Conference on Human Factors in Computing Systems
- **Honorable Mention Award (Top 5% of Submissions)** 2021
ACM International Conference on Interactive Surfaces and Spaces
- **Honorable Mention Award (Top 5% of Submissions)** 2021
IEEE Visualization Conference

ACADEMIC SERVICE

- **Program Committee** 2026
ACM CHI Conference on Human Factors in Computing Systems
- **Program Committee** 2025, 2026
ACM International Conference on Interactive Surfaces and Spaces
- **Program Committee** 2025
IEEE Visualization Conference
- **Student Volunteer** 2022
IEEE Conference on Virtual Reality and 3D User Interfaces
- **Student Volunteer** 2020
Australian Conference on Human-Computer Interaction

REVIEWING

- CHI, ISS 2026
- CHI, ISMAR, ISS, VIS, VR · TVCG 2025

• CHI, EuroVis, ISMAR, ISS, MobileHCI, SIGGRAPH, VIS, VR, VRST · IJHCI, TVCG	2024
• ISMAR, ISS, UIST, VIS, VR, VRST · Frontiers, IJHCI, JCSS, TVCG	2023
• CHI, ISMAR, MobileHCI, VIS, VR	2022
• CHI, EuroVis, ISMAR, ISS, VIS	2021
• VIS	2020
• CHI	2019

SUPERVISION

Doctoral Students

• Co-advisor of Tobias Rau <i>Topic: AR for Computational Chemistry</i> <i>With Prof. Michael Sedlmair</i>	Since 2024
• Co-advisor of Nina Doerr <i>Topic: Visual Highlighting in the Real World</i> <i>With Prof. Michael Sedlmair and Prof. Dieter Schmalstieg</i>	Since 2023
• Co-advisor of Carlos Quijano-Chavez <i>Topic: Interaction Techniques for Situated Analytics</i> <i>With Prof. Dieter Schmalstieg and Prof. Michael Sedlmair</i>	Since 2023
• Co-advisor of Xingyao Yu <i>Thesis Title: On-Body Visualization with Extended Reality: Exploration and Application</i> <i>With Prof. Michael Sedlmair</i>	2023 – 2025

Master's Students

• Co-advisor of Xiaoyan Hu <i>Thesis Title: Collaborative Data Visualisation in Virtual Reality</i> <i>With Prof. Tim Dwyer</i>	2019
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Bachelor's Students

• Advisor of Vivien Schraitle <i>Thesis Title: Supporting Cross-reality Transfer of Virtual Objects in Hybrid Desktop and Augmented Reality Environments</i>	2023 – 2024
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TEACHING

• Head Teaching Assistant at Monash University <i>FTI5147 - Data Visualisation and Exploration</i>	Semester 2 2021 & Semester 1 2022
• Teaching Assistant at Monash University <i>FIT3146 - Maker Lab</i>	Semester 2 2020
• Teaching Assistant at Monash University <i>FTI5147 - Data Visualisation and Exploration</i>	Semester 1 2020 & Semester 1 2021

INVITED TALKS

- "Web-based Immersive Analytics with Anu.js" Spring 2025
New York University
- "Immersive Analytics: Data Visualization and Exploration using Immersive Technologies" Fall 2024
New York University
- "Immersive and Situated Analytics" June 2023
University of Stuttgart
- "Immersive and Situated Analytics" May 2023
Graz University of Technology